

KANKAANPAA NIINISALO PUOLVOIM, Finland

WMO: 029420

Lat: 61.839N

Lon: 22.463E

Elev: 125

StdP: 99.83

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-23.0	-19.8	-25.5	0.4	-22.8	-22.0	0.5	-19.6	8.1	-2.0	7.1	-2.0	1.7	80	0.608

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.4	26.8	18.1	24.9	17.1	23.0	16.1	19.7	24.4	18.4	22.7	17.3	21.4	3.1	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.1	13.2	21.7	16.8	12.1	20.3	15.6	11.2	18.9	56.7	24.6	52.6	22.8	49.0	21.7	24.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.2	6.3	5.6	DB	-25.7	28.9	3.8	2.2	-28.5	30.5	-30.7	31.8	-32.8	33.0	-35.6	34.6	(4)
(5)			WB	-25.9	21.0	3.9	1.8	-28.7	22.3	-31.0	23.4	-33.1	24.4	-35.9	25.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	4.6	-6.2	-6.3	-2.6	3.4	9.2	13.7	16.8	15.3	10.1	4.4	-0.1	-3.5	(6)	
	DBStd	9.22	6.10	6.43	4.66	3.55	4.28	3.56	3.21	3.13	3.41	4.01	4.47	5.94	(7)	
	HDD10.0	2565	502	458	389	200	67	7	0	1	40	178	303	419	(8)	
	HDD18.3	5062	761	691	648	448	286	147	69	102	246	433	553	678	(9)	
	CDD10.0	584	0	0	0	2	42	116	211	167	44	3	0	0	(10)	
	CDD18.3	39	0	0	0	0	1	6	22	9	0	0	0	0	(11)	
	CDH23.3	360	0	0	0	0	21	72	199	68	0	0	0	0	(12)	
	CDH26.7	61	0	0	0	0	1	10	43	7	0	0	0	0	(13)	
(14) Wind	WSAvg	2.8	2.6	2.8	3.0	2.8	2.9	2.8	2.6	2.6	2.6	2.8	2.8	2.8	(14)	
(15) Precipitation	PrecAvg	628	44	33	35	35	37	58	76	77	68	61	57	48	(15)	
	PrecMax	881	89	94	66	85	89	138	175	179	142	117	132	99	(16)	
	PrecMin	456	10	2	1	4	5	6	3	12	11	20	7	11	(17)	
	PrecStd	105	21	21	16	21	23	32	39	37	33	27	28	21	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.2	4.7	10.2	18.2	25.8	27.9	30.2	27.5	20.9	13.8	9.7	6.9	(19)	
		MCWB	3.8	2.9	4.9	10.4	15.2	17.6	20.0	19.0	15.9	11.2	9.0	6.3	(20)	
	2%	DB	2.8	3.1	6.9	14.9	22.9	25.4	27.7	25.2	19.0	12.2	8.0	5.4	(21)	
		MCWB	2.3	2.2	3.4	8.8	13.9	16.5	19.0	17.9	14.7	10.9	7.4	4.9	(22)	
	5%	DB	1.8	1.9	4.9	12.5	20.1	23.0	25.6	23.2	17.1	11.2	6.8	4.2	(23)	
		MCWB	1.4	1.3	2.2	7.1	12.6	15.1	17.9	16.9	13.5	10.1	6.4	3.7	(24)	
	10%	DB	0.7	0.8	3.1	10.1	17.2	20.9	23.3	21.4	15.5	10.1	5.7	3.0	(25)	
		MCWB	0.4	0.4	1.2	5.8	10.7	14.1	17.0	16.1	13.0	9.1	5.3	2.6	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.0	3.4	5.7	11.3	16.7	19.5	21.6	21.0	16.7	12.2	9.1	6.5	(27)	
		MCDB	4.1	4.1	9.3	16.5	22.9	25.3	27.5	25.2	19.6	12.9	9.6	6.7	(28)	
	2%	WB	2.4	2.3	3.9	9.3	14.9	17.6	20.2	19.1	15.4	11.2	7.5	5.1	(29)	
		MCDB	2.6	2.9	6.1	13.8	20.7	22.8	25.3	23.0	17.7	11.9	7.9	5.4	(30)	
	5%	WB	1.5	1.4	2.6	7.7	13.2	16.2	19.0	17.9	14.3	10.3	6.4	3.8	(31)	
		MCDB	1.7	1.9	4.3	11.4	18.9	21.1	23.5	21.9	16.3	10.9	6.7	4.1	(32)	
	10%	WB	0.5	0.5	1.6	6.2	11.5	15.0	17.9	16.8	13.3	9.1	5.3	2.6	(33)	
		MCDB	0.7	0.8	2.8	9.8	16.3	19.9	22.2	20.2	15.2	9.8	5.6	2.9	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	5.2	5.8	7.7	8.8	10.3	9.8	9.4	9.0	7.8	5.4	3.9	4.6	(35)	
		MCDBR	4.1	4.8	9.4	13.4	14.2	13.5	12.8	11.8	9.9	6.1	4.2	4.2	(36)	
	5% WB	MCWBR	4.1	4.3	6.2	7.8	7.0	6.4	5.7	5.6	5.9	4.8	4.0	4.0	(37)	
		MCWBR	3.9	4.6	7.9	11.9	12.7	11.5	10.8	10.0	8.2	5.2	4.0	3.8	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.250	0.298	0.321	0.360	0.356	0.351	0.376	0.362	0.338	0.320	0.270	0.351	(40)	
		taud	2.104	2.207	2.295	2.343	2.424	2.464	2.413	2.445	2.504	2.404	2.178	2.577	(41)	
	Edn at Noon	Edn at Noon	441	660	789	828	865	876	839	821	778	642	411	282	(42)	
		Edh at Noon	36	66	86	101	101	99	101	90	71	55	33	22	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	0.20	0.77	2.26	3.68	5.01	5.37	5.04	3.92	2.33	0.97	0.25	0.11	(44)	
	RadStd	RadStd	0.04	0.17	0.27	0.44	0.53	0.43	0.50	0.42	0.28	0.19	0.05	0.01	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (46 neighbors)	+0.46	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page